

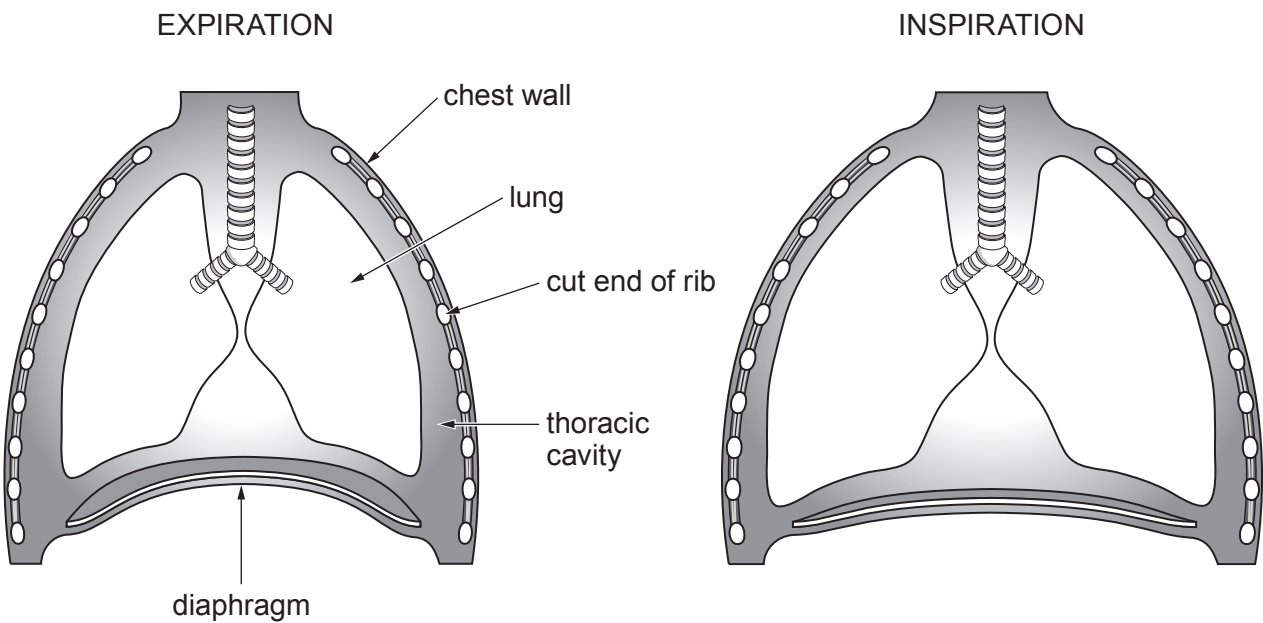
3. The thoracic cavity is the space in the body that contains the lungs. The volume of this space in a healthy adult man is about 5 500 cm³. The lungs occupy 70% of the thoracic cavity. This is around 3 850 cm³.

(a) State **two** other structures which occupy the remaining 30% of the space of the thoracic cavity. [2]

I.

II.

(b) The diagrams below show front views of the thorax during expiration and inspiration.



Use the information in the diagrams above, and your own knowledge, to explain how **inspiration** occurs. [5]

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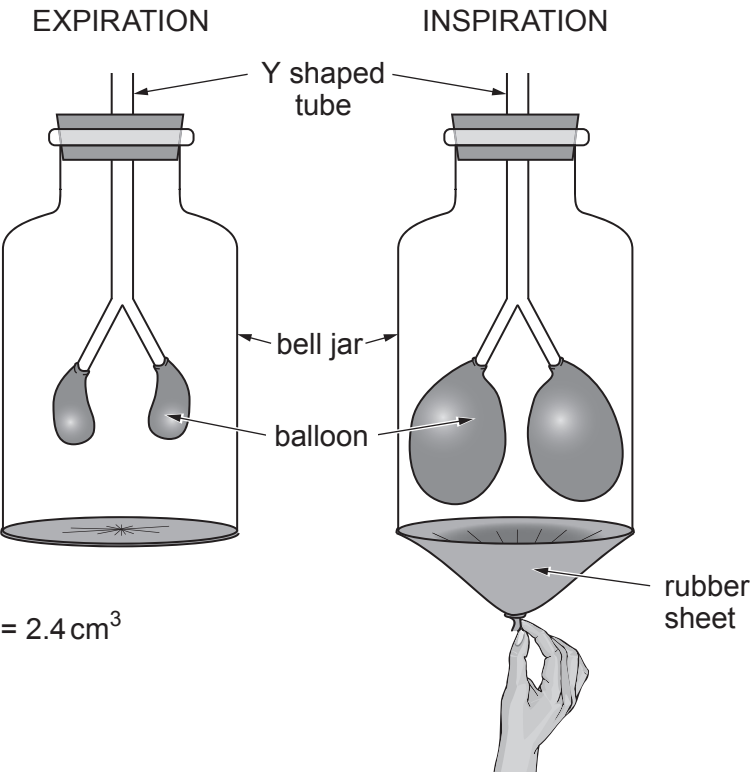
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(c) The bell jar model can be used to demonstrate expiration and inspiration as shown below.



Volume of one deflated balloon = 2.4 cm^3

Volume of bell jar = $5\,500\text{ cm}^3$

- (i) Calculate the percentage of the volume of the bell jar taken up by the **two** deflated balloons. Give your answer to **two decimal places**. [2]

% of volume of bell jar taken up by deflated balloons =

- (ii) Using **only** the information and diagrams given and the calculation above, describe **two** limitations of the bell jar model compared with the human thorax. [2]

I.

II.

